TAEKYUNG KI

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RESEARCH INTERESTS

I am interested in generative models and their applications to creating real-world impact for human. My current research focuses on **controllable, interactive, and reliable video generation** toward **the broader goal of video-based world models**. My previous work spans human video generation, including audio-driven avatar generation for video dubbing [C2], diverse real-time controllability [C4], user-interactive head avatar generation [C6], NeRF-based parametric head reenactment [C3], and training-free methods for controllable and reliable video diffusion sampling [C5, C7].

EDUCATION

Ph.D. in Artificial Intelligence, KAIST *Sep. 2025 - Present*
- Advisor: Prof. Sung Ju Hwang Seoul, South Korea

M.S. in Mathematics, Yonsei University *Mar. 2019 - Feb. 2021*
- Thesis: Deep Scattering Transform with Max-pooling Seoul, South Korea
- Advisor: Prof. Youngmi Hur

B.S. in Mathematics, Kookmin University *Mar. 2015 - Feb. 2019*
- Ranked 1st in department of mathematics. Seoul, South Korea
- Advisor: Prof. Won-Kwang Park

PUBLICATIONS

*: Equal contribution / †: Corresponding author / C: Conference publication / W: Workshop publication / P: Preprint

[C8] **HINT-SD: Targeted Hindsight Self-Distillation for Long-Horizon Agents**
Woongyeng Yeo*, Yumin Choi*, Taekyung Ki, and Sung Ju Hwang
Empirical Methods in Natural Language Processing **(EMNLP Findings)** 2026

[C7] **Self-Refining Video Sampling**
Sangwon Jang*, Taekyung Ki*, Jaehyeong Jo*, Saining Xie, Jaehong Yoon†, and Sung Ju Hwang†
International Conference on Machine Learning **(ICML)** 2026

[C6] **Avatar Forcing: Real-Time Interactive Head Avatar Generation for Natural Conversation**
Taekyung Ki*, Sangwon Jang*, Jaehyeong Jo, Jaehong Yoon, and Sung Ju Hwang†
Conference on Computer Vision and Pattern Recognition **(CVPR)** 2026

[C5] **Frame Guidance: Training-Free Guidance for Frame-Level Control in Video Diffusion Model**
Sangwon Jang*, Taekyung Ki*, Jaehyeong Jo, Jaehong Yoon, Soo Ye Kim, Zhe Lin, and Sung Ju Hwang†
International Conference on Learning Representations **(ICLR)** 2026

[C4] **FLOAT: Generative Motion Latent Flow Matching for Audio-driven Talking Portrait**
Taekyung Ki, Dongchan Min, and Gyeongsu Chae
International Conference on Computer Vision **(ICCV)** 2025
Audio-Visual Generation & Learning Workshop at ICCV (**AVGenL @ ICCV**), 2025

[C3] **Learning to Generate Conditional Tri-plane for 3D-aware Expression Controllable Portrait Animation**

Taekyung Ki, Dongchan Min, and Gyeongsu Chae

European Conference on Computer Vision **(ECCV)** 2024

[C2] **StyleLipSync: Style-based Personalized Lip-sync Video Generation**

Taekyung Ki* and Dongchan Min*

International Conference on Computer Vision **(ICCV)** 2023

[C1] **Deep Scattering Network with Max-pooling**

Taekyung Ki and Youngmi Hur†

IEEE Data Compression Conference **(DCC)** 2021

EXPERIENCE

APPLE *June 2026 - Sep. 2026*
Ph.D. ML Research Intern, Video Computer Vision (VCV) Team Sunnyvale, CA, United States
Research on multimodal diffusion models (Manager: Sikandar Mashayak)

KAIST *Mar. 2025 - Aug. 2025*
Visiting Researcher, MLAI Lab. Seoul, South Korea
Research on video diffusion models (Advisor: Prof. Sung Ju Hwang)

DeepBrain AI Inc. *Dec. 2023 - Mar. 2025*
Research Scientist (Alternative Military Service, South Korea) Seoul, South Korea
Developing novel frameworks for human video generation

AITRICS *Mar. 2022 - Dec. 2023*
Research Scientist (Alternative Military Service, South Korea) Seoul, South Korea
Working with Prof. Sung Ju Hwang and Prof. Eunho Yang
Developing novel frameworks for human video generation

JENTI Inc. *Feb. 2021 - Dec. 2021*
Research Engineer Seoul, South Korea
OCR and NLP-based document analysis

Wavelets and Approximation Lab. at Yonsei University *Aug. 2019 - Feb. 2021*
Graduate Student Researcher (Advisor: Prof. Youngmi Hur) Seoul, South Korea
Mathematical analysis on deep neural networks

Scientific Computing Lab. at Kookmin University *Mar. 2018 - Feb. 2019*
Undergraduate Student Researcher (Advisor: Prof. Won-Kwang Park) Seoul, South Korea
Analysis (PDEs)

TALKS

Motion Latent Flow Matching for Real-time Audio-driven Talking Portrait *Aug. 2025*
at Pika Labs (Host: Chenlin Meng, CTO) Remote

TEACHING

Teaching Assistant at Yonsei University *Mar. 2019 - Dec. 2020*
Calculus 1-2, Linear algebra, Mathematics and biology, and Engineering mathematics (Vector calculus, Fourier analysis, Linear PDEs and statistics, etc.)

AWARDS AND SCHOLARSHIPS

1st Prize Winner at AI Grand Challenge 2021, Ministry of Science and ICT *Oct. 2021*
BK 21 Scholarship, Department of Mathematics, Yonsei University *Sep. 2020 - Feb. 2021*
Top Honors in Mathematics, Department of Mathematics, Kookmin University *Feb. 2019*
Academic Scholarship, Department of Mathematics, Kookmin University *Sep. 2015 - Feb. 2019*

ACADEMIC SERVICES

Conference Reviewer

· **CVPR** (Computer Vision and Pattern Recognition) *2025 -*
· **ICCV** (International Conference on Computer Vision) *2025 -*
· **ECCV** (European Conference on Computer Vision) *2026 -*
· **NeurIPS** (Neural Information Processing Systems) *2026 -*
· **AAAI** (Association for the Advancement of Artificial Intelligence) *2027 -*
· **SIGGRAPH Asia** (Special Interest Group on Computer Graphics and Interactive Techniques) *2026 -*
· **BMVC** (British Machine Vision Conference) *2026 -*
· **ACL ARR** (Association for Computational Linguistics Rolling Review) *2026 -*

REFERENCES

Prof. Sung Ju Hwang Associate Professor, Graduate School of AI, KAIST
- E-mail: sungju.hwang@kaist.ac.kr

Prof. Youngmi Hur Full Professor, Department of Mathematics, Yonsei University
- E-mail: yhur@yonsei.ac.kr

Prof. Won-Kwang Park Full Professor, Department of Mathematics, Kookmin University
- E-mail: parkwk@kookmin.ac.kr